Test day 测试日

A test is only as good as how it is done: test every sample the same way, record every result, and repeat to see how much results vary.

- I can carry out a strength (or melting) test safely, in exactly the same way for every sample.
- I can record repeat results with units and notice anything unusual.

strength 强度 · **load 负载** · **snap 断裂** · **volume 体积** · **millilitre (ml) 毫升** · **mass 质量** · **repeat 重复** · **anomaly 异常值** · **catch tray 接水盘**

THE LESSON

Do Now	3	Samples, folders out; Safety Officer reads hazards, Recorder copies codes, others build 2 rigs.
Hook	2	Plain or fibre sample: which holds more water, and how will each break? Hands-up prediction.
Learn	6	Demo rig (10 cm span, 25 ml pours, 5 s wait), recording rule, roles, safety. Melt test starts.
Do	22	2 rigs per group, samples in mixed order; record ml and how it broke; 3 repeats (min 2).
Think and Reflect	7	Results onto class board; discuss 'Anomaly or not?'; exit ticket; tidy, mop, wash hands.

Hands-on: Groups break-test Week 7 strips on 2 water-bag rigs. Plan A: up to 2 groups melt-test (15 min). Plan B: dough.

Collect or assess: Results table with repeats and notes (Rubric strand 3); group results on class board (photo for Week 9); exit ticket.

SAFETY

- Bag falls on snap: bucket under each bag; hands, feet clear; pourer kneels to side.
- Wet floor (biggest risk): towels on tables, mop by door; mop spills and pause group.
- Plan A: gloves for ice. Plan B: never eat dough; flour allergy: pour or record only.

WATCH OUT

- Same test each time: 10 cm span, centre loop, gentle 25 ml pours, no pressing; Plan A: test within 1 min.
- Record every result, even early snaps, with how it broke; push for 3 repeats (accept 2).

Language: Chinese talk welcome; write in English (Chinese notes count). Traps: load is not loud, snap = break, volume = liquid.

Next week: Week 9 pools class data: photograph the board, keep results sheets in folders, note groups missing data.

PREP FOR ONE CLASS

Plan A: before the lesson about 129 min (105 could be done by someone else); after about 27 min (some times estimated)

Plan B: before the lesson about 111 min (85 could be done by someone else); after about 27 min (some times estimated)

2 OR MORE WEEKS BEFORE

- [ ] Keep 2 spare trial strips for the Hook (slides show 1 plain, 1 fibre) 2 min · teacher

THE WEEK BEFORE

- [ ] Book printing: worksheet x24 (3 sheets each, double-sided, stapled) Plan A 10 min · reprographics
- [ ] Book printing: worksheet x24 (2 sheets each); check exit ticket cut first Plan B 10 min · reprographics
- [ ] Collect rig kit for 6 groups and demo (books, bags, buckets, jugs, towels) 30 min · technician
- [ ] Collect cool boxes, gloves, bags, melting kit; laminate 2 station cards Plan A 20 min · technician
- [ ] If cool boxes are short, bubble-wrap 6 lidded boxes optional Plan A 15 min · technician
- [ ] Tie 13 strings, mark 12 cups at 25 ml; print kit card and rig diagrams 32 min · technician

THE DAY BEFORE

- [ ] Check handover note and board (plan, melting groups, trial); fill folders 10 min · teacher
- [ ] Check dough strips feel hard; check the flour allergy list Plan B 7 min · teacher

ON THE DAY, BEFORE THE LESSON

- [ ] Set out 6 kit trays (2 rigs each, books beside), mop and bucket by door 15 min · technician
- [ ] Set up the demo rig at the front (both plans) 5 min · teacher
- [ ] Just before: move ice strip trays from freezer into insulated boxes Plan A 5 min · teacher

AFTER THE LESSON

- [ ] Photograph class board; check results sheets in folders, read exit tickets 12 min · teacher
- [ ] Clean up: dry books, rinse buckets and jugs, dry towels, mop, return kit 15 min · technician

PRINT

- Worksheet, Plan A x24 (3 sheets each, double-sided, stapled)
- Worksheet, Plan B x24 (2 sheets each, double-sided, stapled)
- Rig diagram A4 x6 (1 sheet each, laminated)
- Rig diagram A3 x1 (board copy)
- Melting station card A5 x2 (laminated, Plan A)
- Kit card for your plan, A5 double-sided (1, or 1 per group)

GET OR MAKE (NEW THIS LESSON)

- Textbooks x78 (students' own are fine)
- Carrier or freezer bags with handles x13
- Buckets or deep bowls x13 (catch trays)
- String x13 lengths (pre-tied hanger strings)
- Plastic cups x12, 25 ml cylinder, marker (pour cups)
- Jugs (about 2 litres) x6, waste bowls x6
- Rulers (30 cm) x7
- Towels x13, paper towels (1 roll), mop and bucket
- Tap water, about 12 litres
- Laminating pouches A4 x6, A5 x2 (A5 for Plan A)
- Plan A: cool boxes with lids x6
- Plan A: thin plastic bags x12, gloves x18 pairs
- Plan A: 10 ml cylinder x2, lids x20, stopwatch x2
- Plan A: lidded boxes x6 + bubble wrap (optional)

Have ready from earlier weeks: Ice strip trays x6 (Plan A, freezer), Dough strip trays x6 (Plan B, dry shelf), 2 spare trial strips (Hook), Group folders + planning frames x6, Role card sets x6, Sentence frame card sets x6, Class project board, Kit trays x6

Chinese: draft, needs native-speaker check